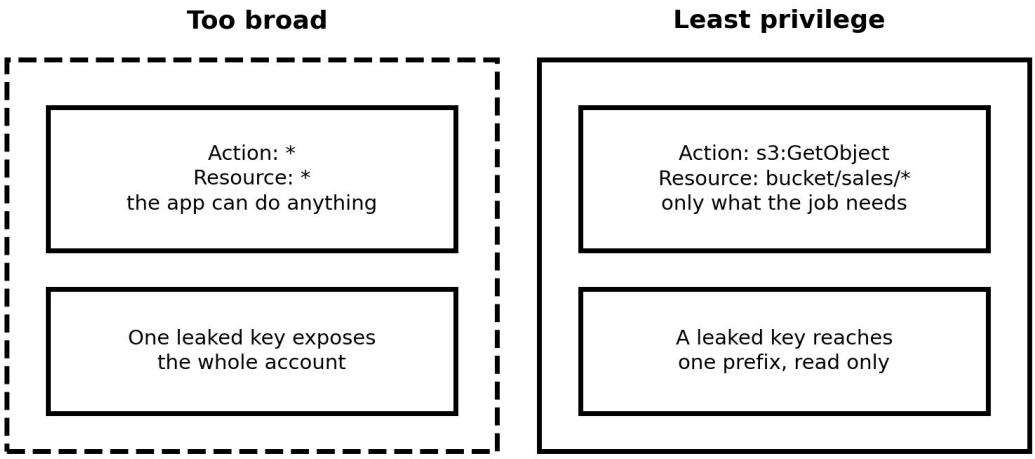


Diagram 3: A broad policy and a least-privilege policy, and what a leaked key can reach in each.

Note. The first rule works on the bucket itself and the second works on the objects inside it. This is why two different ARNs appear. The condition keeps the listing inside the sales/ prefix.

Step	What to do
1. Start small	Begin with no permissions, or with a narrow managed policy, and add only what a real task needs.
2. Write a scoped policy	Replace the broad managed policy with a customer managed policy that names exact actions and ARNs.
3. Let AWS suggest	IAM Access Analyzer can build a policy from the activity recorded in CloudTrail, so the policy matches what was actually used.
4. Remove what is unused	Check the last accessed information of each user and role, and delete the permissions and the access keys that are not used.
5. Add guardrails	Use permissions boundaries and SCPs so that no one can grant more than the organization allows.
6. Review often	Permissions grow over time, so review them on a schedule, for example every quarter.

AWS I & II – Storage, Security, and Pipeline Services

- **Topic:** AWS I & II – Storage, Security, and Pipeline Services
- **Date:** 30th October 2026

3. Amazon S3: Buckets, Objects, and Prefixes

Amazon S3 (Simple Storage Service) is object storage. You store any file as an object inside a bucket and reach it through a web API, a console, or the command line. It is used for data lakes, backups, logs, website files, and the input and output of many other AWS services. S3 is designed for very high durability, 99.999999999% (eleven nines).

Term	Simple meaning
Bucket	The top-level container for objects. The name must be unique across all of AWS, and the bucket is created in one Region, where the data stays unless you copy it.
Object	One stored item: the data itself plus its metadata. An object can be up to 5 TB.
Key	The full name of an object inside its bucket, such as sales/2026/10/orders.csv.
Prefix	The leading part of a key that groups objects, such as sales/2026/10/. It works like a folder in the console, but it is only text.
Version ID	With versioning on, each change to an object gets a version ID, so older versions can be recovered.
Storage class	How the object is stored and billed, from S3 Standard for frequent use to the Glacier classes for archives.

One bucket, with prefixes that look like folders

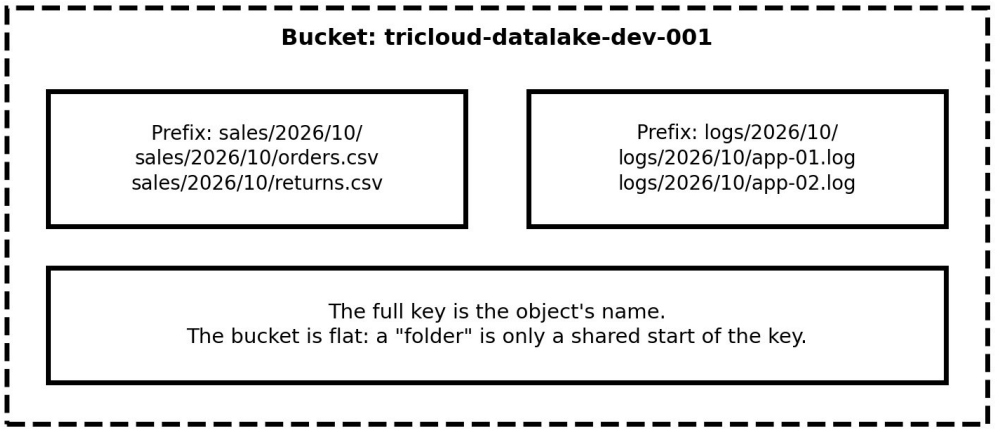


Diagram 4: One flat bucket holds objects, and prefixes only group them by the start of their keys.

Prefixes are not folders. S3 has no real directories. The console shows folders because it splits keys at the "/" character. This means that "renaming a folder" is really copying every object to a new key and deleting the old ones.

Point	ADLS Gen2 (Azure) vs. S3 (AWS)
Top container	ADLS Gen2: a container inside a storage account. S3: the bucket, which has a globally unique name.
Folders	ADLS Gen2: real directories, when the hierarchical namespace is on. S3: only prefixes inside the keys.
Rename or move	ADLS Gen2: one fast operation on the directory. S3: copy each object and then delete the original.
Access control	ADLS Gen2: Azure RBAC roles and ACLs. S3: IAM policies, bucket policies, and Block Public Access.
Storage class	**Best for**
S3 Standard	Data used often, such as the active tables of a data lake.
S3 Intelligent-Tiering	Data with unknown or changing use. S3 moves it between tiers automatically.
S3 Standard-IA and One Zone-IA	Data used rarely but needed quickly. One Zone-IA keeps a single copy in one availability zone.
S3 Glacier classes	Archives. Retrieval takes from minutes to hours, and Deep Archive is the cheapest.

Controlling access. S3 can be reached only by identities that are allowed. Several tools work together.

Tool	Simple meaning
IAM policy	Attached to a user or role. It says which buckets and prefixes the identity may use.
Bucket policy	A resource-based policy on the bucket. It names who may use it, including roles from other accounts.
Block Public Access	A switch that stops any public access to the bucket. It is on by default for new buckets, and it should stay on.
ACLs	The older way of setting permissions on objects. New buckets have them turned off, and policies are used instead.
Pre-signed URL	A temporary link, made by someone who has access, that lets another person download or upload one object for a limited time.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
```

```
    "Sid": "AllowEtlRoleRead",
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::111122223333:role/tricloud-etl-role" },
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::tricloud-datalake-dev-001/sales/*"
   }
  ]
}
```

Step	What to do
1. Create the bucket	In the S3 console choose Create bucket, enter a unique name such as tricloud-datalake-dev-001, and pick the Region.
2. Keep it private	Leave Block Public Access turned on.
3. Turn on versioning	Enable it if overwritten or deleted data must be recoverable.
4. Check encryption	New objects are encrypted by default. Keep the default, or choose a KMS key (section 4).
5. Create and test	Upload a file, then list the prefix to see it.

4. AWS Secrets Manager and KMS Fundamentals

In simple words. AWS KMS (Key Management Service) creates and guards the cryptographic keys that protect data. AWS Secrets Manager stores secrets, such as database passwords and API keys, and hands them to applications that are allowed to read them. Secrets Manager uses KMS to keep each secret encrypted.

KMS term	Simple meaning
KMS key	The key used to encrypt and decrypt. It is created inside KMS, and the key material never leaves KMS in plain form. The default type is a symmetric key.
Alias	A friendly name for a key, such as alias/tricloud-data-key, so that code does not depend on the long key ID.
Key policy	The policy attached to the key. Every key has one, and it controls who may use and manage the key. IAM policies work for a key only if its key policy allows it.
Data key	A small key that KMS creates to encrypt the actual data. It is stored next to

	the data in an encrypted form.
Envelope encryption	KMS encrypts the data key, and the data key encrypts the data. This way big files are encrypted quickly without sending them to KMS.
Key rotation	KMS can replace the key material every year automatically, with no change needed in the applications.
Who owns the key	**Simple meaning**
AWS owned key	Used by AWS inside its own services. It is free, and it is not visible in your account.
AWS managed key	Created by AWS in your account for a service, such as aws/s3 or aws/secretsmanager. You can see it, but you cannot change its policy.
Customer managed key	Created and controlled by you. You write its policy, choose the rotation, and can disable it. It is billed per month and per request.

Secrets Manager. A secret is a value, often a small JSON document, that a person or application needs to sign in somewhere. Secrets Manager stores it encrypted, and an application reads it with an API call at the moment it is needed, so the password never sits in code or in a file.

Reading a secret that is protected by a KMS key



The role needs permission for the secret, and permission to use the key

Diagram 5: An application with a role asks Secrets Manager, which uses a KMS key to decrypt the secret.

Feature	**Simple meaning**
Automatic rotation	Secrets Manager can change the secret on a schedule using a Lambda function. It has ready-made support for databases such as Amazon RDS.
Versions and labels	A secret keeps versions marked AWSCURRENT, AWSPREVIOUS, and AWSPENDING, so a rotation can happen with no break for applications.

Access control	An IAM policy with secretsmanager:GetSecretValue, and optionally a resource policy on the secret.
Audit	Every read and change is recorded in AWS CloudTrail.
Replication	A secret can be copied to other Regions for disaster recovery.
Point	**Secrets Manager vs. KMS**
What it holds	Secrets Manager: secret values such as passwords and API keys. KMS: cryptographic keys that encrypt and decrypt, whose material never leaves KMS.
Main calls	Secrets Manager: GetSecretValue and PutSecretValue. KMS: Encrypt, Decrypt, and GenerateDataKey.
Rotation	Secrets Manager: rotates the secret value on a schedule. KMS: rotates the key material, once a year when it is turned on.
Azure equivalent	Secrets Manager: Key Vault secrets. KMS: Key Vault keys. Certificates go to AWS Certificate Manager.

Note. The secret ARN ends in a random six-character suffix, which is why the policy uses a "-*" at the end. The kms:Decrypt rule is needed only when the secret uses a customer managed key, and the key policy must allow the role as well.

S3 encryption	**Simple meaning**
SSE-S3	S3 manages the keys. It is the default for every new object, and it costs nothing extra.
SSE-KMS	A KMS key encrypts the objects, so key use is logged and can be controlled. Readers need kms:Decrypt as well as s3:GetObject.
SSE-C	The caller supplies the key with every request. AWS does not store it.